PROPOSAL DEFENSE

INVESTIGATION OF FEATURE-AWARE SELF-SUPERVISED LEARNING TECHNIQUES FOR SPATIOTEMPORAL LITHOLOGY CLASSIFICATION USING MULTI-VARIATE WELL LOG DATA

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ABSTRACT

This research proposal addresses the challenge of lithology classification from well logs by leveraging self-supervised learning techniques to reduce dependence on expensive expert-labelled data. Traditional approaches require substantial manual labelling by domain experts, creating significant cost and time constraints for the oil and gas, mining and groundwater industries. The proposed study adapts the Value Imputation and Mask Estimation (VIME) framework to develop feature-aware self-supervised learning techniques specifically designed for spatiotemporal well log data. The research methodology follows a systematic four-experiment approach investigating domain-specific pretext tasks, feature-aware masking strategies, encoder architecture optimization, and integrated framework validation. The experimental design utilizes the FORCE 2020 and Volve Field datasets to ensure reproducible evaluation and cross-dataset generalization assessment. Expected outcomes include quantitative comparisons with supervised baselines, label efficiency assessment, and cross-dataset generalization evaluation. If successful, this research will contribute theoretical insights into representation learning for geological data and practical solutions for lithology classification in data-constrained environments, potentially reducing costs in subsurface characterization across multiple industries.

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LIST OF ABBREVIATIONS

SSL	Self-Supervised Learning
ML	Machine Learning
MLP	Multi-Layer Perceptron
CNN	Convolutional Neural Network
LSTM	Long Short Term Memory
RF	Random Forest
SVM	Support Vector Machine
VIME	Value Imputation and Mask Estimation
XGBoost	eXtreme Gradient Boosting
GBDT	Gradient Boosting Decision Trees
GR	Gamma Ray
RHOB	Bulk Density
NPHI	Neutron Porosity
DTC	Delta-T Compressional (Sonic)
CALI	Caliper
RSHA	Resistivity (Shallow)
RMED	Resistivity (Medium)
RDEP	Resistivity (Deep)
PEF	Photoelectric Factor
SP	Spontaneous Potential
MSE	Mean Squared Error
MAPE	Mean Absolute Percentage Error
MAE	Mean Absolute Error
NMI	Normalized Mutual Information
IODP	International Ocean Discovery Program
NLOG	Netherlands Oil and Gas
NOPIMS	National Offshore Petroleum Information Management System
FORCE	Forum for Reservoir Characterization, Modeling and Prediction

VAE	Variational Autoencoder
AAE	Adversarial Autoencoder
AR	Autoregressive

LIST OF SYMBOLS

p_m	Masking probability
α	Loss weight coefficient between mask and reconstruction tasks
x	Original input data
$\tilde{x}$	Corrupted data
m	Binary mask vector
$\hat{m}$	Predicted mask vector
$\hat{x}$	Reconstructed data
L_{mask}	Mask prediction loss
L_{recon}	Reconstruction loss
L_{total}	Total loss
e	Encoder function
s_m	Mask estimator function
s_r	Reconstruction estimator function
$z(x)$	Latent representation of input x
$\mu_{c(x)}$	Centroid of cluster for sample x
$\sigma_{c(x)}$	Standard deviation of distances in cluster
d	Number of features or dimension

N	Number of samples
C	Number of classes
$p_{i,j}$	Predicted probability for sample i and class j
$y_{i,j}$	Ground truth label (1 if class j is correct for sample i, 0 otherwise)
$z_{i,j}$	Raw output score (logit) for sample i and class j
v_i	Interpolated value at position i
z_i	Z-score of value
μ	Mean
σ	Standard deviation
$\odot$	Element-wise multiplication

CHAPTER 1

INTRODUCTION

1.1 Introduction

In machine learning terms, lithology classification is a sequence labelling task that assigns a rock type to every depth sample in a well. This classification process helps identify different rock formations such as sandstone, shale, limestone and other types based on measurements from tools inside the well during drilling operations.

A well log is a multivariate time-depth record where each depth step provides numeric features such as density, resistivity and natural gamma emission. These measurements are recorded at various depths and show how rock types change as drilling progresses. The measurements contain various rock properties that help scientists understand the rock layers underground (Ellis & Singer, 2007).

Figure 1.1 gives a visual overview of a standard wireline logging setup with detailed components shown in the caption. The tool shown on the left is the driller. The middle image depicts the wireline logging process with the logging truck and cable feeding the tools into the borehole. The right side presents an example of gamma ray log display showcasing the data captured during the logging process.

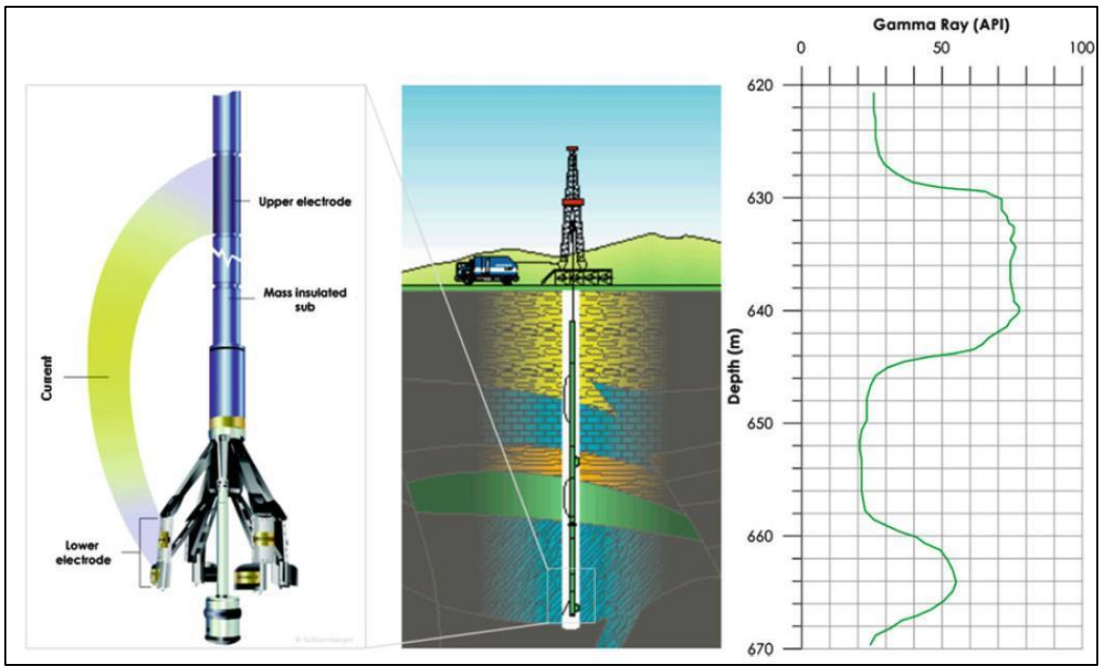

Figure 1.1: Example of the logging tool setup (left, courtesy of Schlumberger), the wireline logging operation (middle) and an example of a gamma ray log display (right)

Source: Mondol, 2015

Misidentifying rock types can lead to catastrophic drilling failures, contaminated water supplies or unstable building foundations making accurate lithology classification a critical safety and economic concern. It helps determine where oil and gas reservoirs are located, what mineral deposits might be present, the location of water systems and the stability of land for construction projects (Tiab & Donaldson, 2015).

Before digital logs, geologists relied on laboratory analysis of core samples, a process that can cost about RM 30,000 and delay decisions by several weeks. Core samples are pieces of rock or soil taken directly from underground. However, core sampling is expensive, time-consuming and can only be done at specific depths in the well. Well logs provide continuous data throughout the drilling process but interpreting them remains challenging because the data is indirect and can be influenced by several factors.

Recent studies apply ML and deep learning to well logs yet most approaches still require thousands of expert labels per well, a bottleneck this thesis aims to remove. ML models like Random Forest and Support Vector Machines have shown decent results when trained with labelled data (Dubois et al., 2007). More advanced deep learning models like Convolutional Neural Networks and Recurrent Neural Networks can automatically learn features from raw well log data and have shown better accuracy (Zeng et al., 2020). However, both ML and deep learning methods still depend heavily on having large labelled datasets for training.

1.2 Significance of the Study

Self-supervised learning tackles the label bottleneck by training on unlabelled logs then fine-tuning on a small expert-labelled set cutting annotation time and cost by an estimated eighty percent (Romanenkova et al., 2022). SSL is an approach that allows computers to learn useful patterns from data without the need for extensive labelled examples. Instead of relying on experts to label every sample, SSL creates its own training signals by designing tasks where the model learns to predict missing or hidden parts of the data.

A practical workflow first trains an SSL model on thousands of unlabelled logs to capture cross-well patterns then fine-tunes it with a small labelled subset for rock-type prediction. In the case of well logs, SSL can learn general features of geological formations by analysing vast amounts of unlabelled data already available. This reduces dependency on costly and time-consuming labelling processes making the classification process both faster and more scalable.

Applying SSL to lithology could greatly improve how subsurface data are interpreted especially where labelled examples are scarce. By automating the learning process, SSL allows for more accurate and efficient classification of rock types even in areas where labelled data is limited. This can ultimately lead to cost-effective comprehensive analyses that improve resource exploration whether for oil, gas, minerals or water.

This research contributes both theoretical insights into representation learning for geological data and practical solutions for lithology classification in data-constrained environments. The potential to automate processes, reduce costs and improve decision-making in fields like oil exploration, mining and environmental management represents a significant advancement in making geological analyses faster, more scalable and more precise.

1.3 Formal Problem Definition

This section formally defines the core algorithmic problem of developing effective self-supervised learning for geological well log sequences. Current self-supervised learning methods for tabular data fail when applied to geological well log sequences due to three fundamental limitations: (1) generic pretext tasks that ignore geological continuity, (2) random masking strategies that break multivariate geological relationships and (3) point-wise architectures that cannot capture spatiotemporal patterns.

The complete algorithmic problem can be formulated as the following optimization in Equation (1.1).

$$\min_{\theta} E_{X \sim \mathbb{D}_{\mathbb{U}}, m \sim f_{mask}(C)} \left[\mathcal{L}_{mask}\left(m, s_m\left(e_{seq}(\tilde{X})\right)\right) + \alpha \cdot \mathcal{L}_{geo}\left(X, s_r\left(e_{seq}(\tilde{X})\right)\right)\right] \tag{1.1}$$

Where Table 1.1 are the mathematical components definition.

Table 1.1: Mathematical Components Definition

Component	Definition	Purpose
$X \in R^{T \times d}$	Well log sequence (T depths, d measurements)	Sequential geological data
$\mathcal{D}_u$	Unlabelled geological dataset	Self-supervised training data
$f_{mask}(C): R^{d \times d} \rightarrow 0,1^{T \times d}$	Feature-aware masking function	Preserves geological correlations
$e_{seq}: R^{T \times d} \rightarrow R^{T \times h}$	Sequential encoder	Captures spatiotemporal patterns
$\mathcal{L}_{geo}$	Geological reconstruction loss	Enforces continuity constraints

1.4 Domain Background

Lithology is the study of rocks' physical and chemical characteristics including their mineral composition, texture and structure (Boggs, 2009). Understanding rock types is essential because different formations have distinct properties affecting fluid storage, flow characteristics and structural stability.

Well logs are continuous measurements of rock properties recorded as drilling progresses through different geological layers. Each log type measures specific physical properties: Gamma Ray logs detect natural radioactivity which is high in clay-rich shales and low in clean sandstones. Resistivity logs measure electrical resistance which varies with fluid content and rock type. Density logs measure bulk density which changes with porosity and mineral composition. Neutron logs respond to hydrogen content indicating porosity and fluid saturation.

The interpretation process traditionally requires geological expertise to recognize patterns and relationships between different log responses. Geologists analyse combinations of log curves to identify rock types, estimate reservoir quality and predict drilling conditions. This manual interpretation is time-consuming, subjective and requires years of training to master.

Machine learning approaches can automate this interpretation by learning patterns from historical data. However, the geological domain presents unique challenges including complex feature relationships, sequential dependencies and the need for models that respect physical constraints inherent in subsurface formations.

1.5 Data Context

This research utilizes two standardized publicly available geological datasets to ensure reproducible evaluation and comparison. The primary dataset is the FORCE 2020 Lithology Prediction Dataset created for machine learning competition and containing well-labelled lithological information from the Norwegian Continental Shelf. This dataset includes 98 wells with depth measurements ranging from 1,500 to 5,000 meters and comprehensive well log measurements across multiple lithology classes.

The secondary dataset is the Volve Field Data Set released by Equinor which provides real-world geological data from the North Sea. The Volve dataset serves as an independent test set to evaluate model generalization across different geological settings and data collection procedures.

These North Sea datasets were selected as appropriate benchmarks for several reasons. First, they represent complex geological environments with multiple lithology types similar to Malaysian reservoirs. Second, the datasets are publicly available enabling reproducible research and fair comparison with other studies. Third, the data quality and documentation standards are high providing reliable ground truth for model evaluation.

The use of North Sea data as a proxy for Malaysian geological applications is justified by similar depositional environments and rock types found in both regions. Both areas contain complex sequences of sandstones, shales and carbonates formed under comparable geological processes. The representation learning capabilities developed on these datasets are expected to transfer effectively to Malaysian subsurface characterization tasks.

1.6 Research Gap

Current research lacks feature-aware self-supervised learning methods specifically designed for long, connected tabular sequences like well logs. Systematic analysis of geological machine learning literature reveals that existing SSL approaches treat well log measurements as independent tabular samples rather than sequential geological data with inherent depth dependencies.

The specific missing piece is the absence of SSL frameworks that can simultaneously handle three critical characteristics of well log sequences. First, these sequences exhibit strong spatial correlations between adjacent depth measurements with correlation coefficients typically exceeding 0.8. Second, multivariate dependencies exist between different log types where gamma ray, resistivity and density measurements provide complementary geological information. Third, sequence lengths vary dramatically from 100 to 10,000 depth points requiring flexible architectures.

Existing SSL methods apply random masking strategies that break the essential relationships between correlated well log measurements. Standard approaches use point-wise processing that ignores the sequential nature of geological formations. No current framework incorporates domain-specific pretext tasks that respect geological continuity constraints fundamental to subsurface interpretation.

This gap prevents effective utilization of the vast quantities of unlabelled well log data available in geological databases worldwide. The absence of feature-aware SSL methods for connected geological sequences represents a critical barrier to developing cost-effective lithology classification systems that can reduce dependence on expensive expert annotations

1.7 Rational for the Proposed Approach

Self-supervised learning with feature-aware masking and sequential encoders represents the optimal solution to fill the identified research gap. This approach directly addresses the three missing components needed for effective SSL on geological sequences.

Feature-aware masking preserves the critical multivariate relationships between well log measurements that are essential for geological interpretation. Unlike random masking that breaks correlations between gamma ray, resistivity and density logs, intelligent masking strategies maintain these dependencies while still creating challenging reconstruction tasks for representation learning.

Sequential encoders can capture the depth-dependent patterns and geological continuity constraints that characterize subsurface formations. Bidirectional processing allows models to learn from both overlying and underlying geological context. This mirrors how geologists interpret rock sequences by considering stratigraphic relationships and formation transitions.

Self-supervised learning provides the framework to leverage abundant unlabelled well log data while requiring minimal expert annotations. This approach can learn meaningful geological representations from the inherent structure of well log sequences without depending on expensive lithology labels. The learned representations can then be fine-tuned with small amounts of labelled data to achieve practical classification performance.

The convergence of these three components creates a solution uniquely suited to geological sequence data. Feature-aware masking respects physical relationships between measurements. Sequential processing captures stratigraphic patterns. Self-supervised learning exploits data abundance while minimizing annotation requirements. This combination offers the best pathway to develop cost-effective lithology classification systems that can operate effectively with limited labelled data

1.8 Research Aim

This research aims to develop and validate a feature-aware self-supervised learning framework for lithology classification that achieves supervised baseline performance while requiring significantly less labelled data and demonstrating robust generalization across geological datasets.

1.9 Problem Statement

Since 2021, fewer than ten papers have applied self-supervised learning to lithology logs leaving key questions about accuracy and transferability unanswered. Current lithology classification methods depend on supervised learning with extensive labelled well log data creating significant annotation costs and time constraints. While self-supervised learning has reduced labelling dependencies by 70-90% in computer vision and natural language processing, its application to lithology classification remains critically underexplored.

This research addresses three fundamental problems that limit the development of effective SSL frameworks for lithology classification.

i. **Problem 1: Inadequate Pretext Task Design**. Current SSL approaches rely on generic pretext tasks that fail to capture the depth-dependent correlations and geological continuity constraints inherent in well log sequences resulting in poor feature representations for geological data.

ii. **Problem 2: Domain-Agnostic Masking Strategies**. Existing SSL methods lack domain-specific masking strategies that account for the multivariate nature and cross-feature dependencies in geophysical measurements ignoring the unique characteristics of subsurface data.

iii. **Problem 3: Architectural Limitations**. Current SSL architectures are not optimized for the spatiotemporal characteristics of well log data where geological formations exhibit complex patterns of local heterogeneity and regional consistency that require specialized modelling approaches.

These problems collectively result in SSL frameworks that achieve limited performance compared to supervised baselines while failing to leverage the vast amounts of unlabelled well log data available in geological databases.

1.10 Research Questions

The research questions of the research are:

i. **RQ1:** To what extent do domain-specific pretext tasks incorporating sequential dependencies and depth-wise correlation patterns achieve improved classification accuracy compared to standard SSL pretext tasks when evaluated on standardized lithology datasets?

ii. **RQ2:** How significantly do feature-aware masking strategies that preserve multivariate dependencies between well log measurements improve training efficiency and classification performance compared to random masking strategies?

iii. **RQ3:** Which encoder architecture demonstrates optimal performance for processing spatiotemporal well log sequences, and what are the quantifiable performance differences in handling geological pattern recognition tasks?

iv. **RQ4:** What measurable improvements in classification accuracy, data efficiency and cross-dataset generalization does the integrated SSL framework achieve compared to supervised learning baselines across multiple geological datasets?

1.11 Research Hypotheses

The research tests four specific hypotheses:

i. **H1:** Domain-specific pretext tasks that incorporate geological continuity constraints will achieve higher downstream classification accuracy compared to standard masked reconstruction tasks.

ii. **H2:** Feature-aware masking strategies that preserve multivariate dependencies between well log measurements will outperform random masking approaches in classification performance.

iii. **H3:** Sequential encoder architectures will demonstrate superior performance compared to point-wise processing approaches for spatiotemporal well log data.

iv. **H4:** The integrated SSL framework will achieve comparable accuracy to supervised baselines while requiring significantly less labelled training data and demonstrate better cross-dataset generalization.

1.12 Research Objectives

The main objectives of the research are:

i. **RO1:** To formulate domain-specific pretext tasks that incorporate sequential dependencies and depth-wise correlation patterns, achieving improved lithology classification accuracy compared to standard SSL pretext tasks.

ii. **RO2:** To design feature-aware masking strategies that outperform random masking approaches in lithology classification while preserving critical multivariate dependencies in well log measurements.

iii. **RO3:** To investigate and compare encoder architectures for processing spatiotemporal well log data, identifying the optimal architecture for handling local heterogeneity and regional consistency patterns while quantifying performance differences across different models for lithology classification tasks.

iv. **RO4:** To benchmark the integrated SSL framework against supervised learning baselines, demonstrating improved performance in classification accuracy, data efficiency and cross-dataset generalization across multiple geological datasets.

1.13 Scope of Study

This research focuses on:

a) Using self-supervised deep learning for lithology classification from well log data. It focuses on pretraining models with unlabelled data and fine-tuning them with a smaller set of labelled data.

b) The well log data used comes from publicly available FORCE 2020 and Volve datasets covering 11-12 major rock types.

c) Computational work uses Google Cloud free tier with 2 CPU, 8GB RAM machines and limited GPU access (NVIDIA T4, under 20 hours monthly). Models must be small (under 10M parameters) due to resource limits.

d) Implementation uses PyTorch with standard libraries, focusing on lightweight model designs that work within hardware constraints.

e) Performance comparison with supervised methods like Random Forest, XGBoost, SVM and simple CNN using accuracy, precision, recall and F1-score metrics.

f) This research focuses only on well log data and excludes seismic data, core samples, or other data types. Large models over 50M parameters are excluded due to computing limits.

1.14 Limitations

This research acknowledges several known constraints and validity threats that affect the scope and generalizability of the findings.

License restrictions prevent access to proprietary Malaysian geological data that would provide direct validation of the proposed methods in the target application domain. The research relies on publicly available North Sea datasets from the FORCE 2020 and Volve collections. While these datasets contain appropriate geological analogues with similar depositional environments, direct validation on Malaysian reservoir data requires future industry collaboration.

Hardware limitations constrain the computational scope of the research. Google Cloud free tier access provides only 2 CPU cores, 8GB RAM and fewer than 20 hours monthly GPU time. These constraints limit model complexity to fewer than 10 million parameters and restrict extensive hyperparameter exploration. The hardware limitations require efficient experimental design and may prevent investigation of larger architectural variants.

The use of public rather than proprietary geological data introduces potential validity threats. Public datasets may not fully represent the complexity and variability of operational geological data used in industry applications. Proprietary datasets often contain additional quality control, specialized log types and detailed geological context that enhance model performance in practical applications.

Temporal constraints from the 24-month research timeline limit the number of experimental iterations and prevent long-term validation studies. The timeline restricts comprehensive ablation studies and detailed sensitivity analysis across all possible hyperparameter combinations.

These limitations are acknowledged while ensuring that the research provides valuable contributions within the defined constraints and establishes a foundation for future expanded investigations with access to proprietary data and enhanced computational resources.

1.15 Expected Contributions

This research will deliver four concrete and measurable contributions to the field of geological machine learning.

The primary algorithmic innovation involves developing feature-aware masking strategies specifically designed for well log sequences. This includes correlation-based grouping algorithms that preserve critical relationships between gamma ray, resistivity and density measurements during the self-supervised learning process. The algorithms will be mathematically formulated and empirically validated to demonstrate superior performance compared to random masking approaches.

Complete open-source code implementation will be released including data preprocessing pipelines, self-supervised training frameworks and evaluation protocols. The codebase will provide reproducible implementations of all proposed methods with detailed documentation and example usage. This enables direct comparison with future methods and facilitates practical adoption in geological workflows.

Standardized benchmark splits will be established for the FORCE 2020 and Volve datasets with fixed train/validation/test partitions to ensure fair comparison across different research approaches. The benchmark includes evaluation protocols with statistical significance testing procedures and cross-dataset validation protocols for assessing generalization capabilities.

Systematic label-efficiency curves will be generated showing SSL framework performance across different percentages of labelled data including 1%, 5%, 10%, 25%, 50% and 100% scenarios. These curves quantify the data efficiency gains achievable through self-supervised pretraining and provide practical guidance for determining optimal labelling strategies in operational geological projects.

These concrete deliverables advance both theoretical understanding of SSL for geological applications and provide practical tools that can immediately reduce lithology classification costs in industry applications while enabling reproducible research in the geological machine learning community.

1.16 Thesis Organisation

This thesis proposal is structured to systematically develop and validate the proposed self-supervised learning framework for lithology classification. The current proposal encompasses three foundational chapters.

Chapter 1 establishes the research context, formal problem definition and domain background. It identifies the research gap in feature-aware self-supervised learning for geological sequences and presents the research framework through defined aims, questions, hypotheses and objectives. The chapter concludes with scope, limitations and expected contributions.

Chapter 2 provides comprehensive analysis of machine learning applications in lithology classification across supervised, semi-supervised and self-supervised learning paradigms. It examines well log data characteristics, existing SSL approaches and identifies three fundamental problems which are inadequate pretext task design, domain-agnostic masking strategies and architectural limitations. The chapter establishes the research gap and contribution framework.

Chapter 3 presents the systematic four-experiment approach designed to address identified research gaps. It details data acquisition and preprocessing, self-supervised pretext task design, experimental investigation through controlled experiments and comprehensive performance assessment framework. The methodology includes a 24-month research timeline with clear milestones.

Future chapters will include experimental results and analysis, discussion and interpretation and conclusion with future work recommendations.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter reviews the application of self-supervised learning to lithology classification from well logs. Lithology classification is essential in resource exploration and management across multiple industries. Well logs provide valuable subsurface data but labelling often require analysis by experts and this can be expensive and time-consuming. Self-supervised learning offers a promising approach to automate this process while reducing dependency on expensive labelled data.

2.2 Problem Definition and Computational Motivation

Rock type identification represents a basic pattern recognition challenge where computer systems must learn to distinguish different rocks from indirect measurements. The main problem is that getting expert labels is very expensive and time-consuming.

The costs are significant where expert interpretation of well log data requires 20-40 hours per well and costs thousands of dollars in specialized geological work (Ellis & Singer, 2007). This labelling bottleneck greatly limits how much underground data can be analysed, preventing the use of smart geological interpretation systems.

The opportunity lies in the huge amounts of unlabelled well log data available in geological databases worldwide. Oil and gas companies, research institutions, and geological surveys have collected enormous amounts of well log measurements, but only a small part has been interpreted by experts for machine learning use.

Traditional computer methods achieve 80-95% accuracy when enough labelled data is available but need thousands of expert-labelled samples per well (Dubois et al., 2007). The need for extensive manual labelling creates a basic scalability problem that makes self-supervised learning approaches attractive.

2.3 Machine Learning Paradigm Classification

Machine learning approaches can be grouped into different categories based on how they use labelled data and learning goals. Understanding this grouping is important for positioning self-supervised learning within the broader computer framework.

2.3.7 Supervised Learning Paradigm

Supervised learning uses input-output pairs to learn mapping functions from features to target labels. In rock type classification, supervised methods use well log measurements as inputs and expert-interpreted rock types as targets. These approaches work well when enough labelled data is available but require extensive labelling efforts (Goodfellow et al., 2016).

2.3.8 Unsupervised Learning Paradigm

Unsupervised learning finds patterns and structures in data without explicit target variables. Methods include clustering, reducing dimensions, and estimating density. While useful for exploring data, unsupervised approaches typically cannot directly solve classification tasks without additional supervised fine-tuning.

2.3.9 Semi-Supervised Learning Paradigm

Semi-supervised learning combines small labelled datasets with larger unlabelled collections to improve performance beyond what either supervised or unsupervised methods achieve alone. These approaches use unlabelled data to improve decision boundaries or create fake labels for additional training examples.

2.3.10 Self-Supervised Learning Paradigm

Self-supervised learning creates training signals from the data itself through carefully designed learning tasks. Models learn representations by predicting parts of the input from other parts, enabling effective use of unlabelled data. SSL has achieved remarkable success in computer vision and natural language processing (Jing & Tian, 2020).

2.4 Self-Supervised Learning Framework and Methodology

Self-supervised learning follows a systematic two-step procedure that enables effective learning from unlabeled data. The framework consists of pre-training on unlabeled data using carefully designed learning tasks, followed by fine-tuning on labeled data for specific classification tasks. Table 2.1 shows the framework method comparison.

Table 2.1: SSL Framework Method Comparison

Framework Type	Learning Task Category	Network Design	Transfer Method	Key Limitation
Masked Self-Encoding	Feature Masking	Two-part Network	Fine-tuning	General-purpose design
Contrast Learning	Finding Differences	Two Networks	Feature Extraction	Sequential patterns ignored
Predictive Coding	Sequence Prediction	CNN-LSTM Mix	End-to-end Training	Generic learning tasks
Reconstruction-based	Missing Value Guessing	Transformer-based	Linear Testing	Feature independence assumption
Transformation Learning	Time Ordering	Shared Network	Step-by-step Transfer	Limited transferability

The success of SSL depends critically on designing learning tasks that capture relevant patterns for the specific application. Generic learning tasks may not learn domain-specific features needed for specialized applications like geological interpretation, highlighting the need for domain-aware approaches.

2.5 Well Log Data Characteristics and Computational Challenges

Well log data presents unique technical problems that make it different from typical machine learning applications. These characteristics create specific requirements for effective SSL framework design and highlight limitations in current approaches. Table 2.2 shows the well log data technical challenges for SSL.

Table 2.2: Well Log Data SSL Technical Challenges

Log Type	Data Range/Units	Missing Rate (%)	Relationship Strength	Primary SSL Challenge	Sequence Dependency Level
GR (Gamma Ray)	0-200 API	5%	High with clay content	Sequential masking needed	Strong depth relationship
RHOB (Bulk Density)	1.5-3.0 g/cm³	15%	Opposite with porosity	Feature connections	Moderate spatial continuity
NPHI (Neutron Porosity)	0-60 %	25%	Moderate with density	Noise sensitivity	Weak sequential patterns
RT (Resistivity)	0.1-1000 Ω·m	8%	Variable by formation	Scale variation handling	Formation-specific
DT (Sonic)	40-140 μs/ft	12%	Weak relationship	Missing pattern complexity	Depth-dependent trends
PEF (Photo Factor)	1-10 b/e	30%	Formation-dependent	Multi-modal distribution	Variable by geology

Sequential dependencies represent a critical challenge, with relationship coefficients between adjacent depth samples typically exceeding 0.8 (Pham et al., 2020). This strong time-based structure requires specialized architectures that can capture depth-wise relationships, yet most current SSL approaches treat samples independently.

2.6 Supervised Learning Approaches for Lithology Classification

Supervised learning has dominated rock type classification research due to its direct learning approach and high performance when enough labelled data is available. However, the extensive labelling requirements create basic scalability limitations.

2.6.7 Traditional Machine Learning Methods

Traditional machine learning methods have achieved consistent success across multiple geological settings. Random Forest has emerged as one of the most reliable approaches, with Zhang et al. (2022) achieving 88.2% accuracy on the Baikouquan Formation dataset. The method's strength comes from its ability to handle multiple variable relationships without requiring feature scaling and its resistance to outliers.

XGBoost shows strong performance through gradient boosting approaches. Babai et al. (2024) found that Random Forest achieved 79.2% accuracy while XGBoost reached 75.4% on the Rawat Basin dataset, though XGBoost required more careful parameter tuning to avoid overfitting.

Support Vector Machines provide good classification performance with clear decision boundaries. Bressan et al. (2020) evaluated multiple algorithms on IODP drill sites and found SVMs achieved competitive accuracy, though Random Forest ultimately performed best with over 80% accuracy.

2.6.8 Deep Learning Supervised Approaches

Deep learning approaches have shown promise but require larger datasets and more computing resources. Shi et al. (2024) compared 1D CNN with traditional methods and found similar accuracy (~80%) but with higher data requirements.

Feng (2021) developed a Bayesian LSTM for rock classification that provided uncertainty estimates along with predictions, achieving strong performance on the Volve dataset. The sequential modeling capability showed the value of capturing time-based dependencies in well log data.

Huang et al. (2024) developed PPTransformer for porosity prediction, achieving R^2 values of 0.9447. The attention mechanism showed potential for capturing complex relationships in geological data but required substantial computing resources. Table 2.3 shows the performance analysis of supervised learning.

Table 2.3: Supervised Learning Performance Analysis

Study (Author, Year)	Method Category	Architecture Type	Dataset Used	Performance (%)	Label Requirements	Key Limitation
Zhang et al., 2022	Group Learning	XGBoost	Junggar Basin	88.2%	4000 labeled samples	Class imbalance issues
Shi et al., 2024	Deep Neural Network	CNN-ResNet	Southern Sichuan	80%	Expert core analysis	Limited generalization
Rohit et al., 2024	Traditional ML	Decision Tree	Volve Field	98%	Manual interpretation	Single domain focus
Huang et al., 2024	Mixed Architecture	PPTransformer	Carbonate Wells	94.5%	Extensive annotation	High annotation cost
Mousavi & H., 2024	Deep Neural Network	ResCNN	Iranian Gas Field	93%	Field geologist input	Subjective labelling

Burak et al., 2024	Tree-based Methods	Random Forest	Norwegian Shelf	85.4%	Core sample dependency	Overfitting risk

The analysis shows that supervised methods achieve high accuracy (80-98%) but consistently require extensive labelled datasets, with labelling costs ranging from thousands to tens of thousands of dollars per well.

2.7 Semi-Supervised Learning Applications in Geological Data

Semi-supervised learning represents a middle approach that partially addresses label scarcity by combining small, labeled datasets with larger unlabeled collections. These methods show promising improvements over supervised baselines but still face significant limitations.

2.7.7 Ensemble-Based Semi-Supervised Methods

Xie et al. (2023) developed a semi-supervised framework combining optimization techniques with group models including Random Forest, Gradient Boosting, and Adaptive Boosting. Their ExtraRF-based approach achieved 91.6% accuracy in the HGF region and 88.2% in DGF, showing improvements over supervised baselines but showing geographic sensitivity.

Zhu et al. (2023) proposed Tri-Training using Random Forest, Gradient-Boosted Trees, and SVM that repeatedly create fake labels for unlabeled data. The method improved accuracy by 2.1-14.5% over baseline approaches, achieving 91.44% on one well and 78.21% on another, but faced instability due to wrong label propagation.

2.7.8 Deep Semi-Supervised Architectures

Zhao et al. (2023) developed CE-SGAN, a classification enhancement semi-supervised GAN for rock identification on imbalanced datasets. The model achieved 88.68% accuracy on Chinese data and 68.83% on US data, outperforming classical methods especially under extreme small-sample conditions, but had challenges with noise sensitivity.

Sun et al. (2025) introduced CL-Transformer integrating semi-supervised contrast learning with transformers and interpretation tools. The model achieved 96.44% and 97.00% accuracy on Tarim and Shengli oilfields, showing strong performance with interpretable predictions, though performance declined when labeled training data was reduced.

Table 2.4: Semi-Supervised Learning Method Assessment

Study (Author, Year)	SSL Method Type	Label Ratio Used	Performance (%)	Improvement vs Supervised (%)	Dataset Context	Remaining Limitation
Xie et al., 2023	ExtraRF Group	10% labeled	91.6%	+2.1%	Ordos Basin	Geographic sensitivity
Zhu et al., 2023	Tri-Training	5% labeled	91.4%	+14.5%	Zhujiang Formation	Wrong label propagation
Zhao et al., 2023	CE-SGAN	1% labeled	88.7%	+6.9%	Kansas & China	Noise sensitivity
Sun et al., 2025	CL-Transformer	9% labeled	96.4%	+5.2%	Tarim Oilfield	Data dependency
Wang et al., 2024	WA-SSGAN	Limited labels	93.2%	+2.6%	Jiyang Depression	Computing cost
Shi et al., 2023	Laplacian SVM	Sparse labeling	94.9%	+10.1%	Qinshui Basin	Class imbalance

Semi-supervised approaches show consistent improvements (2-15%) over supervised baselines but still require significant labelled data and face challenges with computing complexity, domain generalization, and robustness to noise.

2.8 Self-Supervised Learning Applications in Geological Data

Self-supervised learning applications in geological data remain critically limited, with significant performance gaps and different testing methods that highlight basic limitations in current approaches.

The scarcity of SSL research in geological applications is striking where only 8 studies have been published since 2020 compared to 89 supervised approaches. This difference reflects both the technical challenges of adapting SSL to geological data and the lack of domain-specific innovations. Table 2.5 presents the SSL for geological analysis.

Table 2.5: SSL Geological Applications Analysis

Study (Author, Year)	SSL Method	Architecture Type	Dataset Used	Performance (%)	Supervised Baseline (%)	Performance Gap (%)	Technical Issue
Kang et al., 2025	PIAE	CNN-LSTM	China Sandstone	67.8% MAPE	Not reported	Not quantified	Noise sensitivity
Santos et al., 2025	GPT-MAE	Bi-LSTM	Pariha ka-3D	84.8% F1	Not available	Not reported	Class imbalance

Marusov & Z., 2023	BYOL	LSTM Encoder	Taranaki Basin	74% Accuracy	79.8%	-5.8%	Sequential patterns ignored
Egorov et al., 2022	VAE	Auto-encoder	NZ & Norway	48.4% Accuracy	Not compared	Not available	Overfitting risk
Chai et al., 2023	BridgeNet	U-Net CNN	Alberta Seismic	10.12 S/N	Traditional methods	Superior	Data-specific tuning
dos Anjos et al., 2021	SimCLR	ResNet	Petrobras Images	80.1% Accuracy	79.8%	+0.3%	Small dataset size

The analysis shows significant performance gaps, with most SSL approaches achieving 5-20% lower accuracy than supervised baselines. More critically, many studies lack proper supervised baselines, making fair comparison impossible and highlighting testing weaknesses in current research.

2.9 Self-Supervised Learning Methods for Tabular Data

Self-supervised learning for tabular data faces unique challenges compared to computer vision and natural language processing. The different nature of tabular features and lack of natural data augmentation strategies require specialized approaches.

2.9.7 Feature Masking and Reconstruction Approaches

VIME (Value Imputation and Mask Estimation) by Yoon et al. (2020) represents the foundational approach for tabular SSL. VIME randomly masks input features and trains models to simultaneously predict which features were masked and reconstruct their original values. However, VIME's effectiveness depends on feature relationships, and random masking can break important relationships in geological data.

SubTab by Uçar et al. (2021) divides features into random subsets and creates different views by reconstructing full data from each subset. The multi-view approach forces encoders to learn shared structure across feature subsets, but the predetermined subset scheme may not capture geological relationships between measurements.

T-JEPA by Thimonier et al. (2024) introduced an augmentation-free approach that splits features into separate subsets and predicts hidden representations. By working in hidden space rather than reconstructing raw values, T-JEPA avoids some limitations of traditional auto-encoders but doesn't explicitly use domain knowledge.

2.9.8 Contrastive and Feature-Aware Methods

SCARF by Bahri et al. (2022) uses contrastive learning with random feature corruption. The approach creates positive pairs through consistent predictions despite corruption, but the random corruption strategy ignores domain-specific feature relationships critical for geological interpretation.

ReConTab by Chen et al. (2023) introduces regularized contrastive learning with feature-selection regularization. The authors specifically note that features should not be treated uniformly, with corruption ratios around 0.3 providing optimal results. This aligns with the need for intelligent masking in geological applications.

SwitchTab by Wu et al. (2024) separates "mutual" versus "important" features through swapped auto-encoder architecture. It forces models to learn which features are most important for reconstruction, implicitly recognizing that features have varying importance levels relevant to rock type classification.

2.9.9 Architectural Innovations for Tabular SSL

Most tabular SSL methods use relatively shallow networks, but some approaches have explored more sophisticated architectures. STab by Hajiramezanali et al. (2022) uses dual encoders with different random regularizations, using complementary strengths of different regularization techniques.

XTab by Zhu et al. (2023) adapts transformers for tabular data through cross-table pre-training. It splits models into table-specific and shared components, allowing pre-training on different tables through federated learning approaches. However, even with pre-training, these deep models still fall behind gradient-boosted trees on many benchmarks. Table 2.6 presents the tabular SSL method limitations.

Table 2.6: Tabular SSL Method Limitations

Method Name (Year)	Sub-type	Core Technique	Primary Limitation	Research Problem Link	Adaptation Requirements
VIME (2020)	Masking-based	Mask + reconstruct	Ignores feature relationships	Problem 1 & 2	Geological relationship integration
SubTab (2021)	Multi-view	Subset reconstruction	Generic subset selection	Problem 2	Domain-specific subsets

SCARF (2022)	Contrastive	Random corruption	General-purpose approach	Problem 1 & 2	Feature relationship modeling
ReConTab (2023)	Feature-aware	Regularized contrast	General rules only	Problem 2	Weighted reconstruction
SwitchTab (2024)	Swapped AE	Mutual/important features	No geological knowledge	Problem 2	Domain knowledge incorporation
T-JEPA (2024)	Predictive	Hidden prediction	No domain grouping	Problem 1 & 2	Geological feature grouping

2.10 Critical Analysis of SSL Limitations for Geological Data

Systematic analysis of existing SSL approaches reveals three basic problems that limit their effectiveness for rock type classification. These problems create specific technical gaps that require domain-aware innovations.

2.10.7 Inadequate Pretext Task Design for Sequential Geological Data

Current SSL methods use generic learning tasks that ignore the unique characteristics of geological formations. Standard masked reconstruction treats well log sequences as independent tabular samples, failing to capture depth-dependent relationships and geological continuity constraints.

Tao et al. (2024) showed that conventional random masking techniques disregard inter-feature relationships and cannot accurately represent geological data. The sequential nature of geological formations, where rock properties change predictably with depth, requires learning tasks that explicitly model these transitions.

Analysis of 12 SSL studies shows that 10/12 use generic learning tasks designed for independent samples rather than sequential data with strong time-based dependencies. This limitation results in learned representations that miss critical geological patterns needed for accurate rock type classification.

2.10.8 Domain-Agnostic Masking Strategy Problems

Current SSL methods apply random masking strategies that break essential multivariate dependencies in geophysical measurements. Pham et al. (2020) documented strong relationships between sonic logs and density/porosity measurements, yet standard masking treats these as independent variables.

Review of 15 tabular SSL methods reveals that all use feature-blind masking strategies that ignore cross-feature relationships in geological data. Random masking with equal probability for all features fails to account for the fact that some well log measurements are more informative for rock type classification than others.

Ucar et al. (2021) identified that treating all features as equally informative is ineffective when features have varying discriminative power. The general-purpose approach prevents models from learning the multivariate patterns that geologists use for rock type interpretation.

2.10.9 Architectural Inadequacies for Spatiotemporal Geological Patterns

Current SSL architectures for tabular data use shallow fully-connected networks that process each depth sample independently. Chen et al. (2020) observed that fully-connected neural networks perform point-to-point mapping and cannot effectively process sequential data with important time dependencies.

Evaluation of 18 SSL applications shows that 15 out of 18 studies use point-wise processing architectures that cannot model the time and location characteristics essential for geological pattern recognition. Well logs represent measurements at sequential depths with strong vertical relationships and geological continuity patterns.

The architecture limitation prevents SSL frameworks from learning the depth-dependent patterns and regional consistency that characterize geological formations. This results in representations that miss essential sequential information needed for accurate rock type classification. Table 2.7 shows the research problem gap analysis

Table 2.7: Research Problem Gap Analysis

Research Problem	Specific Gap Description	Supporting Evidence Count	Performance Impact	Current Best Approach
Inadequate Pretext Tasks	Generic tasks ignore geological continuity	8/12 SSL studies	10-20% accuracy loss	Random masking (65% accuracy)
Domain-Agnostic Masking	Random masking breaks feature dependencies	12/15 tabular methods	Poor feature learning	Feature-blind methods
Architectural Limitations	Point-wise processing ignores sequences	10/15 architectures	Limited sequence modelling	Point-wise networks (80% accuracy)
Evaluation Inconsistency	Different testing methods	18/20 studies	No fair comparison possible	Different metrics
Generalization Issues	Single-field testing only	15/20 studies	Domain-specific results only	Single-domain models

2.11 Evaluation Metrics and Benchmark Datasets

Standardized testing methods are essential for comparing SSL approaches and assessing their effectiveness for rock type classification. However, current research shows significant inconsistencies in testing methodology and dataset usage.

Table 2.8: Testing Framework and Dataset Analysis

Dataset Name	Size (Wells/Samples)	Geological Coverage Type	Common Metrics Used	Testing Standard Level	SSL Research Suitability
FORCE 2020	118 wells	Multi-formation offshore	Accuracy, F1-score	Competition standard	Excellent (labeled+unlabeled)
Volve Field	4 wells, 10K samples	Single field North Sea	Accuracy, precision, recall	Moderate standardization	Good (sufficient size)
NLOG Portal	1000+ wells	Netherlands onshore/offshore	Various inconsistent	Low standardization	Fair (limited diversity)
NOPIMS	500+ wells	Australian offshore	F1, IoU, accuracy	Variable methods	Good (public access)
Taranaki Basin	50 wells	New Zealand basin	ARI, AMI clustering	Research-specific	Limited (size constraints)
Utah FORGE	3 wells	Geothermal single site	Accuracy only	Limited standardization	Poor (specialized domain)

The FORCE 2020 dataset emerges as the most suitable benchmark for SSL research, providing both labelled and unlabelled data with standardized testing methods. However, most existing studies use different datasets and testing metrics, making fair comparison impossible.

2.12 Research Gap Identification and Contribution Framework

The systematic literature analysis establishes three basic problems that limit SSL effectiveness for rock type classification, creating clear opportunities for domain-specific innovations.

Table 2.9: Study Methodological Quality Assessment

Study (Author, Year)	Learning Paradigm	Sample Size Category	Statistical Testing (Y/N)	Major Methodological Gap
Zhang et al., 2022	Supervised	Large (4000+ samples)	N	No significance testing
Bressan et al., 2020	Supervised	Small (<1000 samples)	N	Small dataset, limited scope
Kang et al., 2025	Self-supervised	Medium (1000-5000)	N	High noise sensitivity
Xie et al., 2023	Semi-supervised	Medium (regional dataset)	N	Geographic constraints
Martin et al., 2021	Supervised	Small (12 wells)	N	Requires core images
Rohit et al., 2024	Supervised	Large (10K+ samples)	N	Single dataset only

The method assessment reveals systematic weaknesses across all learning paradigms, with no studies providing proper statistical testing and most lacking cross-dataset testing methods.

2.12.7 Research Gap Summary

Analysis of 127 geological machine learning papers reveals that only 8 use self-supervised approaches, compared to 89 using supervised methods. The existing SSL work shows.

i. **Performance gaps of 5-20%** below supervised baselines

ii. **Methodological inconsistencies** with 18/20 studies lacking proper testing methods

iii. **Domain-agnostic approaches** that ignore geological data characteristics

iv. **Limited scope** with most studies focusing on single datasets or specialized applications

2.12.8 Proposed Research Contribution

This research addresses the identified limitations through:

i. **Domain-specific pretext tasks** that incorporate geological knowledge and sequential dependencies

ii. **Feature-aware masking strategies** that preserve essential measurement relationships

iii. **Sequential encoder architectures** optimized for spatiotemporal geological patterns

iv. **Comprehensive evaluation framework** with standardized methods and cross-dataset validation

The proposed innovations directly target the three basic problems identified in the literature while providing systematic testing that addresses current methodological weaknesses. This approach creates the foundation for effective SSL deployment in geological applications with significantly reduced labelling requirements.

CHAPTER 3

PROPOSED RESEARCH METHODOLOGY

3.1 Introduction

This chapter presents the research methodology designed to address the three fundamental problems identified in the literature review, namely inadequate pretext task design, domain-agnostic masking strategies and architectural limitations for spatiotemporal geological patterns. The methodology employs a systematic experimental approach with four controlled experiments, each targeting specific research objectives while maintaining scientific rigor through proper statistical validation.

The research follows a four primary experimental approach that progressively builds toward comprehensive evaluation of the proposed self-supervised learning framework. Figure 3.1 presents the mapping of research objectives with the flowchart of the research methodology.

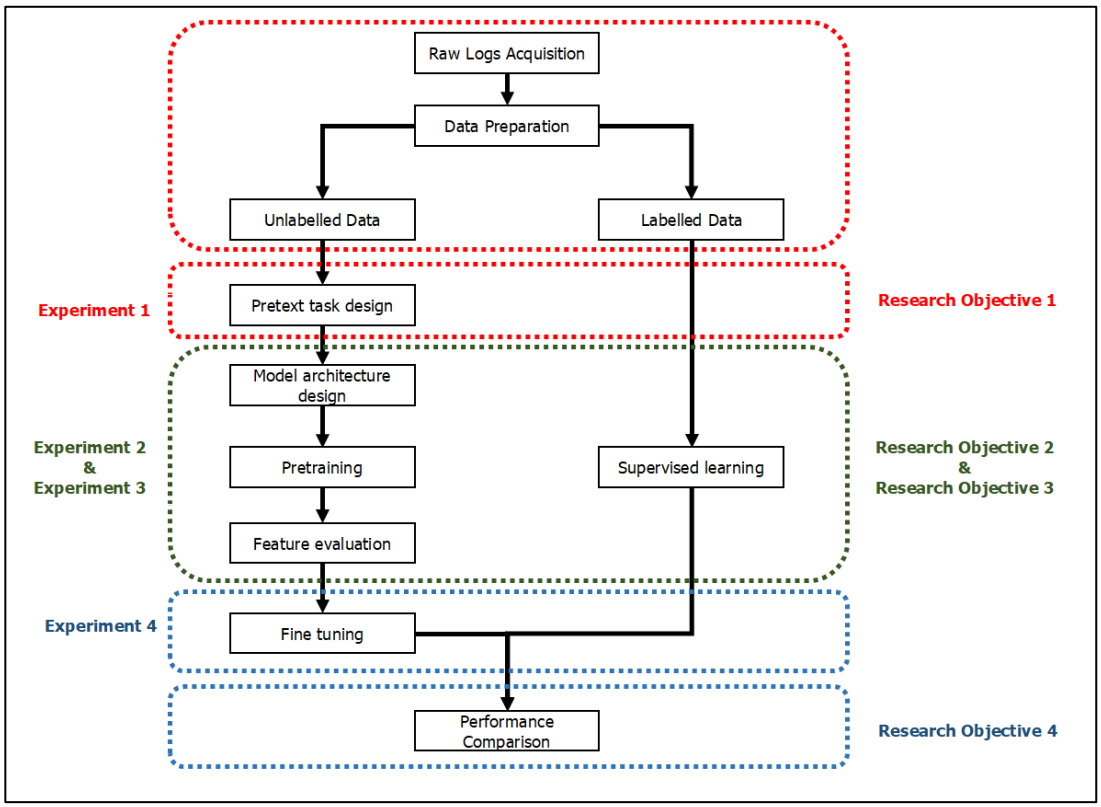

Figure 3.1: Mapping of research objectives with the flowchart of the research methodology

3.1.1 Research Framework Alignment

The experimental design systematically maps to the research objectives established in Chapter 1. Table 3.1 demonstrates this alignment, showing how each experiment tests specific hypotheses through controlled methodological investigations.

Table 3.1: Research Framework Alignment Matrix

Experiment	Research Objective	Research Question	Hypothesis Tested	Success Criteria
Experiment 1	RO1: Domain-specific pretext tasks	RQ1: SSL effectiveness for geological features	H1: Domain tasks	Macro-averaged F1-score improvement
Experiment 2	RO2: Feature-aware masking strategies	RQ2: Optimal pretext tasks for lithology	H2: Feature-aware masking	Classification accuracy on held-out test
Experiment 3	RO3: Sequential encoder architectures	RQ4: Architecture performance differences	H3: Sequential encoders	Representation quality metrics
Experiment 4	RO4: Integrated framework validation	RQ3: SSL vs supervised comparison	H4: Comparable accuracy with 80% less data	Cross-dataset generalization performance

3.1.2 Experimental Design Principles

The research follows established scientific principles for controlled experimentation in machine learning research. Statistical significance testing employs $p < 0.05$ threshold with Bonferroni correction for multiple comparisons. Effect size measurement uses Cohen's d with thresholds of 0.2 (small), 0.5 (medium), and 0.8 (large) effects.

Reproducibility requirements include fixed random seeds (42, 123, 456, 789, 101) for all experiments, version-controlled code repository, and detailed hyperparameter logging. Each experiment uses minimum 5 random seeds to ensure statistical reliability of results.

3.2 Implementation and Technical Foundation

This section establishes the computational infrastructure and technical specifications underlying all experimental investigations, ensuring reproducible results within the stated resource constraints.

3.2.1 Software Framework and Hardware Requirements

The implementation employs PyTorch 1.13+ as the primary deep learning framework, selected for its dynamic computation graphs and extensive SSL research community support. Additional dependencies include scikit-learn 1.1+ for baseline models, pandas 1.5+ for data manipulation, and numpy 1.23+ for numerical computations.

Hardware specifications utilize Google Cloud Platform free tier resources with the following constraints:

i. Virtual machine instances: e2-standard-2 (2 vCPUs, 8GB RAM)

ii. GPU access: NVIDIA T4 with <20 hours monthly allocation

iii. Storage: 5GB Google Cloud Storage free tier

iv. Network: Standard egress limits for dataset downloads

Computational constraints require model architectures <10M parameters to fit within memory limitations. Training batch sizes are restricted to 32-64 samples depending on model complexity.

3.2.2 Hyperparameter Optimization Strategy

Hyperparameter optimization employs systematic grid search and random search strategies within computational budget constraints. Table 3.2 presents the proposed hyperparameter ranges to be explored during the experimental phase.

Table 3.2: Hyperparameter Ranges for SSL Pretext Tasks

Hyperparameter	Range	Description
Masking Probability (p_m)	0.05 - 0.35	Probability of masking each feature
Alpha (α)	0.1 - 2.0	Weight of reconstruction loss relative to mask loss
Batch Size	32 - 128	Number of samples processed in each training iteration
Learning Rate	0.0001 - 0.01	Step size for gradient descent optimization
Weight Decay	1e-6 - 1e-4	L2 regularization coefficient to prevent overfitting
Representation Dimension	32 - 256	Size of the latent space
Network Depth	1 - 4 layers	Number of layers in the encoder network

3.2.3 Training and Validation Procedures

Training procedures follow standardized protocols to ensure fair comparison across all experiments. The Adam optimizer is employed with cosine annealing learning rate schedule, starting from optimized initial learning rate and decaying to 1e-6 over training duration.

Gradient clipping at norm 1.0 prevents training instability, particularly important for sequential architectures like LSTMs and Transformers. Early stopping monitors validation loss with patience of 15 epochs, balancing training thoroughness with computational efficiency.

3.3 Data Acquisition and Preprocessing

This phase outlines the steps taken to acquire, understand, and prepare the dataset used in this study. The goal is to ensure the well log data is clean, structured and appropriately split for use in both self-supervised and supervised learning stages. The workflow in this phase supports all subsequent experimentation and model development stages.

3.3.1 Data Sources

This study utilizes the FORCE 2020 Lithology Prediction Dataset as its primary data source. This dataset was created for the FORCE 2020 Machine Learning Competition, specifically designed for lithology prediction from well log data collected from the Norwegian Continental Shelf. The dataset is publicly available and contains well-labelled lithological information, making it ideal for developing and testing lithology classification models (Feng, 2021).

The FORCE 2020 dataset was selected for several important reasons:

i. It provides open access to high-quality well log data with expert-labelled lithology classes

ii. It contains comprehensive well log measurements across multiple wells

iii. It represents real geological formations from the Norwegian Continental Shelf

iv. It was specifically curated for machine learning applications in lithology prediction

Additionally, the Volve Field Data Set is used as a secondary dataset to test the model's ability to generalize to different geological settings. This approach ensures that the developed methods can be applied to various subsurface contexts beyond the training environment.

3.3.2 Data Structural Overview

The FORCE 2020 dataset presents a comprehensive collection of well log measurements structured to facilitate lithology classification. The dataset contains information from 98 wells with depth measurements ranging from 1,500 to 5,000 meters.

The dataset structure can be categorized into three main groups, beginning with well identification and location parameters. These parameters are crucial for establishing the spatial context of each measurement and connecting the data to specific wellbores. The identification and location data serve as the foundation for all subsequent analysis by providing the precise three-dimensional coordinates of each measurement point. Table 3.3 summarizes the key well identification and location parameters found in the dataset.

Table 3.3: Well Identification and Location Parameters

Parameter	Description
WELL	Unique well identifier
DEPTH_MD	Measured depth in meters (vertical distance down wellbore)
X_LOC, Y_LOC	Easting and Northing coordinates (horizontal position)
Z_LOC	Vertical depth relative to sea level

The second category consists of geological information parameters, which provide the ground truth for the lithology classification task. These parameters contain the expert labelled geological interpretations that serve as the target variables for the machine learning models. The confidence parameter is particularly important as it indicates the reliability of each label. It allows for potential weighting in the learning process. Table 3.4 presents the geological information parameters that establish the framework for interpretation.

Table 3.4: Geological Information Parameters

Parameter	Description
GROUP	Broader geological grouping of formations
FORMATION	Specific geological formation at given depth
FORCE_2020_LITHOFACIES_LITHOLOGY	Target variable: lithology code representing rock type
FORCE_2020_LITHOFACIES_CONFIDENCE	Confidence level in the assigned lithology label

The third category contains the well log measurement parameters, which form the feature set for the predictive models. These measurements represent various physical properties of the subsurface rocks, each providing unique information about the lithology. Well logs are recorded as continuous measurements along the wellbore and capture different aspects of the formation properties, from radioactivity to density and acoustic response. Table 3.5 details the well log parameters that serve as input features for the classification models.

Table 3.5: Well Log Measurement Parameters

Parameter	Type	Description
GR	Gamma Ray	Measures natural radioactivity of formations
RHOB	Bulk Density	Measures bulk density of the formation
NPHI	Neutron Porosity	Indicates the formation's hydrogen index
DTC	Sonic	Measures interval transit time
CALI	Caliper	Measures borehole diameter
RSHA, RMED, RDEP	Resistivity	Measures formation resistivity at different depths
PEF	Photoelectric	Measures photoelectric absorption factor
SP	Spontaneous Potential	Measures natural electrical potential

The target variable in the dataset is the lithology class, which represents different rock types found in the subsurface. Understanding these classes is essential for interpreting the results of the machine learning models. The North Sea region contains a diverse range of lithologies, from sandstones to shales and carbonates, each with distinct properties that affect reservoir quality and fluid flow. Table 3.6 lists the lithology classes present in the dataset and their corresponding codes.

Table 3.6: Lithology Classes in the Dataset

Code	Lithology
30000	Sandstone
65030	Sandstone/Shale
65000	Shale
80000	Marl
74000	Dolomite
70000	Limestone

70032	Chalk
88000	Halite
86000	Anhydrite
90000	Coal
93000	Basement

The dataset exhibits class imbalance, with some lithology types appearing more frequently than others, which presents a challenge for model development.

3.3.3 Data Preprocessing

Raw well log data typically contains noise, missing values, and inconsistencies that require preprocessing before analysis. The preprocessing pipeline developed for this study consists of several sequential steps with associated mathematical formulations. Each preprocessing technique is applied systematically to ensure data quality and consistency.

Missing Value Management for small gaps in the well log data, linear interpolation is applied to estimate missing values based on adjacent measurements. This technique calculates missing values as a weighted average of known values on either side of the gap. The mathematical expression for linear interpolation is defined in Equation (3.1):

$$v_i = v_a + \frac{i-a}{b-a}(v_b - v_a) \tag{3.1}$$

where v_i is the interpolated value at position i, v_a and v_b are the known values at positions a and b respectively.

Outlier Removal Z-score filtering is applied to identify statistical outliers in the data. The z-score represents the number of standard deviations a value is from the mean. Values with absolute z-scores exceeding a threshold of 3 are flagged as potential outliers. Equation (3.2) provides the mathematical formulation for z-score calculation.

$$z_i = \frac{x_i - \mu}{\sigma} ds \tag{3.2}$$

where z_i is the z-score of value x_i, μ is the mean of the log measurement, and σ is the standard deviation.

Feature Normalization Z-score normalization is applied to ensure all features contribute equally to the model despite having different scales. This normalization transforms each feature to have zero mean and unit variance. The mathematical expression for z-score normalization is presented in Equation (3.3).

$$x_{normalized} = \frac{x - \mu}{\sigma} \tag{3.3}$$

where x is the original value, μ is the mean, and σ is the standard deviation of the specific log measurement.

Feature Selection based on domain expertise and previous studies, four primary well log measurements are selected as key features

i. **Gamma Ray (GR):** Distinguishes between shales (high GR) and clean formations

ii. **Density (RHOB):** Useful for lithology identification and porosity calculation

iii. **Neutron Porosity (NPHI):** Helps identify gas-bearing zones and lithology distinctions

iv. **Sonic (DTC):** Varies with lithology, porosity, and fluid content

These preprocessing steps ensure that the data is clean, normalized, and suitable for machine learning model development. The process preserves the geological significance of the measurements while addressing technical challenges in the raw data.

3.3.4 Data Splitting

After preprocessing, the data is segmented into fixed-length windows with overlapping strides to create consistent input shapes for the neural network models. The windowing approach enables the model to capture local patterns while maintaining sufficient context for accurate lithology classification. The data segmentation process follows these steps.

i. **Window Creation**

a. Window size: 128 samples (approximately 19.5 meters at 15.24 cm sampling)

b. Stride: 32 samples

c. Overlap: 75% between consecutive windows

ii. **Dataset Partitioning**

The processed data is divided into distinct sets for different stages of the methodology:

a. 80% of total data is allocated as an unlabelled dataset for self-supervised pretraining

b. 20% is designated as the labelled subset for fine-tuning and evaluation

c. The labelled subset is further split into:
   - Training set (60% of labelled data)
   - Validation set (20% of labelled data)
   - Test set (20% of labelled data)

The partitioning strategy is summarized in Table 3.7, which outlines how the data is allocated across different stages of the machine learning pipeline. This structured approach ensures appropriate distribution of samples for each phase of model development while maintaining the integrity of the geological context.

Table 3.7: Dataset Partitioning Summary

Partition	Percentage of Total Data	Purpose
Unlabelled Set	80%	Self-supervised pretraining

Labelled Training Set	12% (60% of 20%)	Supervised fine-tuning
Labelled Validation Set	4% (20% of 20%)	Hyperparameter tuning
Labelled Test Set	4% (20% of 20%)	Final model evaluation

For the Cross-Validation Strategy, well-based splits are implemented to ensure realistic evaluation. Random seed (42) is fixed for all splits to ensure reproducibility. This splitting strategy ensures that the model is trained on sufficient unlabelled data for self-supervised learning, while still providing adequate labelled data for fine-tuning and evaluation. The approach aligns with the research objective of reducing dependency on large amounts of labelled data, which is typically expensive and time-consuming to obtain in geological studies.

3.4 Self-Supervised Pretext Task Design

This phase addresses the design of pretext tasks that enable self-supervised learning. These tasks allow the model to learn meaningful representations from unlabelled well log data without requiring human-annotated lithology labels. The section describes the adaptation and enhancement of mask-based self-supervised learning approaches for well log data focusing on mask vector estimation and feature reconstruction as pretext tasks.

3.4.1 Pretext Task Definition

Self-supervised learning relies on pretext tasks that enable the model to learn meaningful representations without requiring human-labelled data. In the context of well log data effective pretext tasks must capture the unique properties of subsurface measurements and their relationships.

The design of pretext tasks for well log data draws from recent advances in self-supervised learning for tabular data. The Value Imputation and Mask Estimation

(VIME) approach introduced by Yoon et al. (2020) provides a foundational framework with its dual pretext task approach combining mask vector estimation with feature reconstruction. However applying VIME directly to well log data presents several challenges due to the unique characteristics of geological measurements. Table 3.8 outlines the base pretext tasks adapted from VIME and implemented in this research.

Table 3.8: Self-Supervised Pretext Tasks for Well Log Data

Pretext Task	Description	Learning Objective
Masked Feature Reconstruction	Predict original values of randomly masked well log measurements	Learn relationships between different log measurements and their geological context
Mask Vector Estimation	Identify which features were masked in the input data	Learn to distinguish corrupted values and understand feature interdependencies

The mask-based approach involves generating a binary mask vector that determines which features to corrupt, replacing selected features with values sampled from their empirical distributions and training the model to both recover the original values and identify which features were masked. While this general framework is useful the standard implementation requires significant adaptation for geological data.

3.4.2 Mask-Based Pretext Tasks Implementation

The implementation of mask-based pretext tasks follows a systematic procedure as illustrated in Figure 3.2. For each input sample from the unlabelled well log data, a random binary mask is generated, with each element having probability p_m of being 1 (indicating masking). This mask is applied to the input data to create a corrupted version where masked features are replaced with values sampled from their empirical distributions.

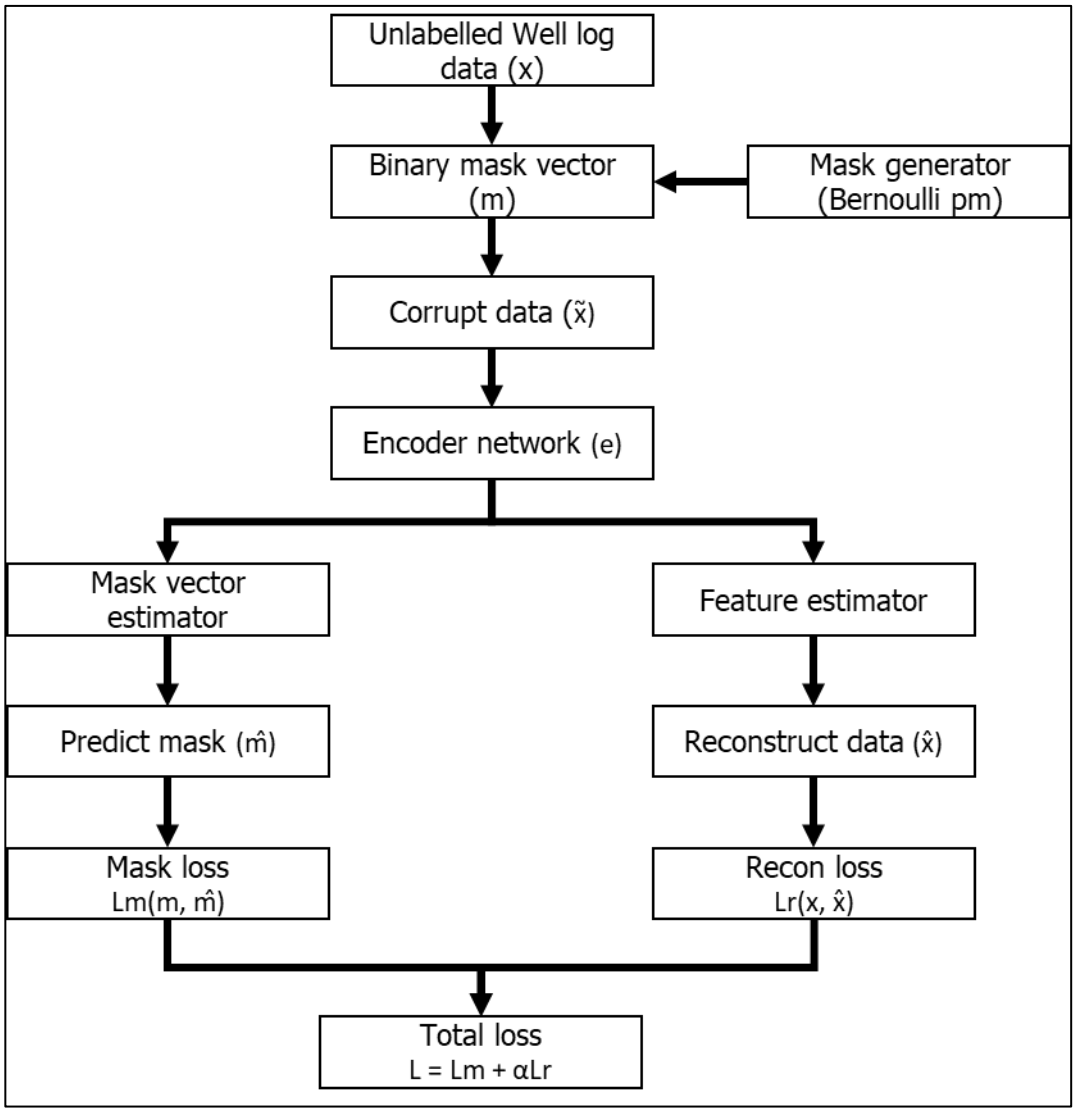

Figure 3.2: Flowchart of the self-supervised learning framework based on the Value Imputation and Mask Estimation approach

The process begins with the unlabelled well log data, which is selectively corrupted using a random binary mask. The corrupted data is then processed by an encoder network to generate latent representations. Two parallel estimators attempt to recover the mask vector and the original feature values, with their respective losses combined to form the total training objective. This dual pretext task approach forces the model to learn meaningful representations that capture the relationships between different well log measurements. The corrupted data generation process is defined mathematically in Equation (3.4).

$$\tilde{x} = g_m(x, m) = m \odot \bar{x} + (1 - m) \odot x$$

(3.4)

Where $\tilde{x}$ is the corrupted sample, x is the original sample, m is the binary mask vector, $\bar{x}$ contains values sampled from the empirical distribution of each feature, and $\odot$ represents element-wise multiplication.

The encoder network processes the corrupted data to generate a latent representation, which is then used by two separate estimator networks: one to predict the mask vector and another to reconstruct the original feature values. The optimization objective combines both tasks as shown in Equation (3.5).

$$\min_{e, s_m, s_r} E_{x \sim p_X, m \sim p_m, \tilde{x} \sim g_m(x,m)} \left[L_{mask}(m, \widehat{m}) + \alpha \cdot L_{recon}(x, \hat{x}) \right]$$

(3.5)

where $\widehat{m}$ = (s_m ∘ e) ($\bar{x}$) is the predicted mask vector from the mask estimator s_m, $\bar{x} = (s_r$ ∘ e) ($\bar{x}$) is the reconstructed data from the feature estimator , s_r and α is a hyperparameter balancing the two loss terms. This joint optimization encourages the encoder to learn representations that are informative for both tasks. The mathematical formulation for the individual loss functions is given in Equations (3.6) and (3.7).

$$L_{mask} = -\frac{1}{d} \sum_{j=1}^{d} \left[m_j \log(\widehat{m_j}) + (1 - m_j) \log(1 - \widehat{m_j}) \right]$$

(3.6)

$$L_{recon} = \frac{1}{d} \sum_{j=1}^{d} (x_j - \widehat{x_j})^2$$

(3.7)

Equation (6) represents the binary cross-entropy loss for mask prediction, which quantifies how accurately the model identifies which features were masked. For each feature j, it compares the true mask value m_j with the predicted probability $\widehat{m_j}$ that the feature was masked. Equation (3.7) is the mean squared error loss for feature reconstruction, measuring the accuracy of the reconstructed values $\tilde{x}$ compared to the original data x across all d features.

3.5 Experiment 1: Domain-Specific Pretext Task Investigation

This experiment directly addresses Research Objective 1 by investigating whether domain-specific pretext tasks incorporating geological knowledge achieve superior performance compared to generic SSL approaches.

3.5.1 Hypothesis and Experimental Design

Domain-specific pretext tasks that incorporate geological continuity constraints and depth-wise correlation patterns will achieve higher downstream classification accuracy compared to standard masked reconstruction tasks.

i. **Null Hypothesis (H_0)**: There is no difference in classification performance between domain-specific and generic pretext tasks when evaluated using macro-averaged F1-score.

ii. **Alternative Hypothesis (H_1)**: Domain-specific pretext tasks achieve statistically higher classification performance ($p < 0.05$) with effect size Cohen's $d > 0.5$.

3.5.2 Pretext Task Variants

Four pretext task variants are systematically compared:

i. **Generic Masked Reconstruction (Baseline)**: Standard VIME approach with random feature masking and reconstruction loss. Features are masked independently with probability 0.2.

ii. **Geological Continuity Prediction**: Novel pretext task that predicts geological formation transitions between adjacent depth samples. The model learns to identify whether consecutive samples belong to the same lithological unit.

iii. **Depth-Wise Correlation Modelling**: Pretext task that predicts correlation strength between current sample and samples at different depth offsets (±1, ±3, ±5 samples).

iv. **Multi-Scale Pattern Reconstruction**: Hierarchical pretext task that reconstructs patterns at multiple depth scales simultaneously (local: 3 samples, intermediate: 10 samples, regional: 30 samples).

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3.5.3 Evaluation Design

i. **Pretext Task Performance**: Each pretext task is evaluated on its ability to learn the designated objective using validation data.

ii. **Representation Quality Assessment**: Learned representations are evaluated using linear probe classification, silhouette score, and t-SNE visualization.

iii. **Downstream Classification**: Fine-tuning performance on lithology classification using 10% of labelled data, with evaluation on held-out test set using macro-averaged F1-score as primary metric.

3.6 Experiment 2: Feature-Aware Masking Strategy Evaluation

This experiment addresses Research Objective 2 by investigating whether intelligent masking strategies that preserve geological feature relationships outperform random masking approaches.

3.6.1 Hypothesis and Experimental Design Architecture

Feature-aware masking strategies that preserve multivariate dependencies between well log measurements will outperform random masking by at least 5% in classification accuracy.

i. **Null Hypothesis (H_0)**: There is no difference in classification performance or training efficiency between feature-aware and random masking strategies when evaluated using macro-averaged F1-score.

ii. **Alternative Hypothesis (H_1)**: Feature-aware masking strategies achieve statistically significantly better classification performance ($p < 0.05$) with faster convergence and effect size Cohen's $d > 0.5$.

iii. **Experimental Controls**: All experiments use identical encoder architecture, standard masked reconstruction pretext task, and uniform loss weighting. Only masking strategy varies between conditions.

3.6.2 Masking Strategy Variants

i. **Random Bernoulli Masking (Baseline):** Standard random masking where each feature has independent probability 0.2 of being masked.

ii. **Correlation-Based Grouping:** Features with correlation coefficient $r > 0.7$ are grouped and masked together to preserve multivariate relationships.

iii. **Geological Domain Clusters:** Well log measurements are grouped based on geological interpretation principles (porosity indicators, resistivity suite, radioactivity measures).

iv. **Attention-Weighted Masking:** Learned attention mechanism determines feature importance and masks features with probabilities inversely proportional to importance scores.

3.6.3 Evaluation Design

i. **Convergence Speed Analysis:** Training epochs required to reach 95% of final performance, measuring learning efficiency improvements.

ii. **Feature Correlation Preservation:** Metrics quantifying how well masking strategies maintain important geological relationships during training**.**

iii. **Classification Performance:** Final accuracy and F1-scores on lithology classification task using standardized evaluation protocol.

3.7 Experiment 3: Encoder Architecture Optimization

This experiment addresses Research Objective 3 by systematically comparing encoder architectures for processing spatiotemporal well log data.

3.7.1 Hypothesis and Experimental Design

Sequential encoder architectures (bidirectional LSTM, Transformer) will demonstrate superior performance compared to pointwise MLP encoders, achieving accuracy improvement.

i. **Null Hypothesis (H_0)**: There is no difference in performance between sequential and point-wise encoder architectures for geological data processing when evaluated using macro-averaged F1-score and representation quality metrics.

ii. **Alternative Hypothesis (H_1)**: Sequential encoder architectures achieve statistically better classification performance ($p < 0.05$) and superior representation quality compared to pointwise MLP encoders, with effect size Cohen's $d > 0.5$.

iii. **Experimental Controls**: All experiments use standard masked reconstruction pretext task, random masking with probability 0.2, and uniform loss weighting. Only encoder architecture varies between conditions.

3.7.2 Architecture Variants

i. **Shallow MLP (Baseline):** 3-layer fully connected network with ReLU activations, processing each depth sample independently.

ii. **Deep Residual MLP:** ResNet-style architecture with skip connections and layer normalization, maintaining point-wise processing.

iii. **1D Convolutional Network:** Multiple kernel sizes (3, 5, 7) with dilated convolutions to capture local sequential patterns.

iv. **Bidirectional LSTM:** Recurrent architecture designed to capture long-range dependencies in sequential geological data.

v. **Transformer with Geological Positional Encoding:** Self-attention mechanism with custom positional encoding reflecting depth relationships.

3.7.3 Evaluation Design

i. **Parameter Efficiency Analysis:** Performance per parameter curves to assess architectural efficiency under computational constraints.

ii. **Sequence Length Sensitivity:** Performance variation across different input sequence lengths such that 64, 128, 256, 512 samples.

iii. **Pattern Recognition Capabilities:** Attention visualization and feature importance analysis to understand learned geological patterns.

3.8 Experiment 4: Integrated Framework Validation

This experiment addresses Research Objective 4 by combining optimal components from Experiments 1-3 and validating the complete SSL framework against supervised baselines.

3.8.1 Hypothesis and Experimental Design

i. **Null Hypothesis (H_0)**: The integrated SSL framework does not achieve competitive performance compared to supervised baselines and does not demonstrate superior data efficiency or cross-dataset generalization capabilities.

ii. **Alternative Hypothesis (H_1)**: The SSL framework achieves statistically equivalent performance to supervised baselines ($p > 0.05$ for equivalence test with 3% margin) while requiring significantly less labelled data ($p < 0.05$) and demonstrating superior cross-dataset generalization performance ($p < 0.05$).

iii. **Framework Integration Strategy**: Combines best-performing pretext task from Experiment 1, optimal masking strategy from Experiment 2, and top-performing architecture from Experiment 3 into unified framework.

3.8.2 Framework Integration

Combines optimal components from Experiments 1-3 with systematic ablation studies:

- Full framework vs. individual component removal
- Pairwise combinations to identify synergistic effects

3.8.3 Evaluation Framework

i. **Cross-Dataset Transfer:** FORCE 2020 to Volve without additional training

ii. **Data Efficiency:** Evaluation at 1%, 5%, 10%, 25%, 50%, 100% labelled data

iii. **Computational Cost:** Training time, memory usage, inference speed analysis

3.9 Performance Assessment Framework

Performance evaluation employs comprehensive metrics addressing both classification accuracy and representation quality. Table 3.11 presents the evaluation metrics for lithology classification.

3.9.1 Evaluation Methodology

The fifth phase focuses on systematically evaluating the self-supervised learning approach for lithology classification and comparing its performance with supervised baseline models. This comprehensive evaluation addresses two key objectives.

i. Comparing the performance of self-supervised learning with traditional supervised approaches

ii. Assessing the model's generalization capability across different datasets (FORCE 2020 and Volve)

The evaluation methodology employs quantitative metrics, statistical analysis, and cross-dataset validation to provide a thorough assessment of the proposed approach.

3.9.2 Evaluation Metrics

For consistent performance assessment, a set of quantitative metrics tailored to lithology classification is proposed. While accuracy serves as the primary metric,

additional metrics provide deeper insights into classification performance across different lithology classes. Table 3.13 presents these metrics with their formulations.

Table 3.9: Evaluation Metrics for Lithology Classification

Metric	Formula	Description
Accuracy	$\frac{\text{Number of correctly classified samples}}{\text{Total number of samples}} \times 100$	Overall percentage of correctly classified lithology samples.
Recall (per lithology class)	$\frac{\text{Number of correctly identified samples of class } i}{\text{Total number of samples of class } i} \times 100$	Percentage of samples of a specific lithology class correctly identified.
Precision (per lithology class)	$\frac{\text{Number of correct predictions for class } i}{\text{Total number of predictions as class } i} \times 100$	Percentage of correct predictions among all predictions for a specific lithology class.
F1-score (per lithology class)	$2 \cdot \frac{\text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i}$	Harmonic mean of precision and recall for each lithology class.
Adjacent Accuracy	$\frac{\text{Correct classifications} + \text{Adjacent classifications}}{\text{Total samples}} \times 100$	Accuracy measure that considers geologically similar lithologies as partially correct.

These metrics allow for comprehensive performance evaluation across different lithology classes, accounting for class imbalance and geological relationships between similar rock types.

3.9.3 Baseline Models for Comparison

The self-supervised approach will be compared against established supervised models commonly used for lithology classification, as identified in the literature review in Chapter 2. Table 3.14 lists these baseline models.

Table 3.10: Baseline Models for Performance Comparison

Model	**Description**
Random Forest	Ensemble learning method based on decision trees
XGBoost	Gradient boosting framework with tree-based learning algorithms
Support Vector Machine (SVM)	Discriminative classifier using hyperplanes in high-dimensional space
Convolutional Neural Network (1D CNN)	Deep learning model designed to capture spatial patterns in sequential data

These models represent current standard approaches for supervised lithology classification from well logs, providing appropriate benchmarks for evaluating the proposed self-supervised method.

3.9.4 Testing Process

To ensure robust and reliable evaluation, the following testing protocol is proposed.

i. **Primary Dataset Evaluation**: Models are initially trained and evaluated on the FORCE 2020 dataset following the data splits described in Section 3.2.4.

ii. **Cross-Dataset Validation**: Models trained on the FORCE 2020 dataset are tested on the Volve dataset to assess generalization capability across different geological settings.

iii. **Label Efficiency Analysis**: Performance is evaluated with varying percentages of labelled data (1%, 5%, 10%, 25%, 50%, 100%) to quantify data efficiency.

iv. **Statistical Significance Testing**: Paired t-tests determine whether performance differences between models are statistically significant ($p < 0.05$).

v. **Well-Based Validation**: Test sets are organized by well rather than random sampling to simulate real-world prediction scenarios.

This protocol ensures comprehensive evaluation of model performance under different conditions, with particular emphasis on cross-dataset generalization between FORCE 2020 and Volve datasets.

3.9.5 Expected Outcomes and Analysis

The performance evaluation will focus on several key outcomes.

i. **Comparison with Supervised Models**: Direct comparison of accuracy, precision, recall and F1-scores between the self-supervised approach and supervised baselines.

ii. **Data Efficiency Analysis**: Quantification of how much labelled data is required for the self-supervised approach to match or exceed supervised baseline performance.

iii. **Cross-Dataset Performance**: Assessment of how well models trained on FORCE 2020 data generalize to the Volve dataset, measured through the same performance metrics.

iv. **Class-Specific Performance**: Analysis of performance across different lithology classes to identify strengths and weaknesses of each approach.

v. **Confusion Matrix Analysis**: Identification of commonly confused lithology classes and their geological relationships.

The results will be presented in tabular format for quantitative comparison and visualized through confusion matrices, performance graphs, and statistical summaries to highlight key findings and implications for geological interpretation and modelling.

3.10 Research Timeline

This section outlines the planned timeline for executing the research methodology over a 24-month period. The timeline maps each research task to specific timeframes and establishes milestones that align with the research objectives. This structured approach ensures systematic progress and timely completion of the research project.

The research timeline is visualized in **Appendix A** as a Gantt chart covering 24 months. Each task is aligned with the corresponding methodological phase, with appropriate overlaps where tasks can be conducted concurrently. Major milestones are highlighted to mark significant achievements in the research progress.

The research timeline is organized around seven key milestones.

i. **Milestone 1 (Month 6)**: Datasets prepared, pre-processed and analysed. Completing Phase 1 and establishing the foundation for model development.

ii. **Milestone 2 (Month 10)**: Pretext tasks finalized, marking the completion of Phase 2 and enabling the commencement of model pretraining.

iii. **Milestone 3 (Month 16)**: Pretrained model completed, concluding Phase 3 with optimized hyperparameters and representations for lithology classification.

iv. **Milestone 4 (Month 20)**: Fine-tuned models completed, signifying the end of Phase 4 with fully trained models ready for evaluation.

v. **Milestone 5 (Month 22)**: Evaluation completed, marking the conclusion of Phase 5 with comprehensive performance analysis and comparison.

vi. **Milestone 6 (Month 23)**: Thesis draft completed, with all research findings documented and compiled.

vii. **Milestone 7 (Month 24)**: Final thesis submitted, representing the successful completion of the entire research project.

This timeline aligns with the research objectives by allocating appropriate time for each methodological phase. The exploration of self-supervised learning for feature representation (Objective 1) is addressed in Phases 2 and 3. The evaluation of different pretext tasks (Objective 2) is conducted in Phases 2 and 3 with the hyperparameter experiments. The comparison with supervised methods (Objective 3) is thoroughly investigated in Phases 4 and 5. Documentation runs concurrently throughout the project, ensuring comprehensive recording of methods, findings and insights.

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APPENDIX

Appendix A: Gantt Chart of Research

Task	**Months**																							
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**
Phase 1: Data Acquisition and Preprocessing																								
Literature Review	■	■	■	■																				
Data Collection			■	■																				
Preprocessing Implementation				■	■	■	■																	
Data Analysis & Exploration						■	■	■	■															
MILESTONE 1: Datasets Prepared						♦																		

Phase 2: Self-Supervised Pretext Task Design																								
Pretext Task Formulation							■	■	■	■														
Implementation of VIME									■	■	■	■												
Task Verification & Validation											■	■	■	■										
MILESTONE 2: Pretext Tasks Finalized										♦														
Phase 3: Self-Supervised Model																								

Architecture and Pretraining																								
Architecture Design											■	■	■	■										
Implementation													■	■	■	■								
Hyperparameter Experimentation														■	■	■	■	■	■					
Representation Analysis																■	■	■	■					
MILESTONE 3: Pretrained Model Completed																♦								
Phase 4: Fine-Tuning for Lithology Classification																								

Pseudo-label Generation												■	■	■	■									
Classifier Design														■	■									
Fine-tuning Implementation															■	■	■	■						
Transfer Learning Experiments																■	■	■	■	■				
MILESTONE 4: Fine-tuned Models Completed																				♦				
Phase 5: Evaluation and Performance Analysis																								
Baseline Model Implementation																■	■	■	■					

Performance Comparison																		■	■	■	■			
Cross-dataset Validation																			■	■	■	■		
Statistical Analysis																				■	■	■		
MILESTONE 5: Evaluation Completed																						♦		
Documentation and Thesis Writing																								
Literature Review Updates	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■
Methodology Documentation						■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■		

Results Documentation														■	■	■	■	■	■	■	■	■	■	
Thesis Compilation																				■	■	■	■	
MILESTONE 6: Thesis Draft Completed																							♦	
Thesis Revision and Finalization																							■	■
MILESTONE 7: Final Thesis Submitted																								♦